

AR(1) : Score Equations - Solutions

$$\ell(\underline{\theta}; \underline{x}) = -\frac{n}{2} \log(2\pi\sigma_z^2) + \frac{1}{2} \log(1-\phi_1^2) \\ - \frac{1}{2\sigma_z^2} S(\underline{x}; \mu, \phi_1)$$

$$\frac{\partial \ell}{\partial \mu} \Big|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0$$

$$\Rightarrow -\frac{1}{2\hat{\sigma}_z^2} \frac{\partial S(\underline{x}; \mu, \phi_1)}{\partial \mu} \Big|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0 \quad (1)$$

$$S(x; \mu, \phi_1) = d_{00} - 2\phi d_{01} + \phi_1^2 d_{11} \\ - 2\mu(1-\phi_1)(t_0 - \phi_1 t_1) \\ + \mu^2(1-\phi_1)(n - \phi_1(n-2))$$

$\therefore \ln (1)$

$$-\frac{1}{2\hat{\sigma}_z^2} \left[-2(1-\hat{\phi}_1)(t_0 - \hat{\phi}_1 t_1) \right. \\ \left. + 2\hat{\mu}(1-\hat{\phi}_1)(n - \hat{\phi}_1(n-2)) \right] = 0$$

$$\Rightarrow \hat{\mu} = \frac{t_0 - \hat{\phi}_1 t_1}{n - \hat{\phi}_1(n-2)}$$

$$\frac{\partial \ell}{\partial \phi_1} \Big|_{\hat{\phi}_1, \hat{\mu}, \hat{\sigma}_z^2} = 0$$

$$\Rightarrow -\frac{\hat{\phi}_1}{1 - \hat{\phi}_1^2} - \frac{1}{2\hat{\sigma}_z^2} \frac{\partial S(\underline{x}, \hat{\mu}, \phi_1)}{\partial \phi_1} \Big|_{\hat{\phi}_1, \hat{\mu}, \hat{\sigma}_z^2} = 0 \quad (2)$$

$$S(\underline{x}, \mu, \phi_1) = \sum_{t=1}^n (x_t - \mu)^2 - 2\phi_1 \sum_{t=2}^n (x_t - \mu)(x_{t-1} - \mu) + \phi_1^2 \sum_{t=2}^{n-1} (x_t - \mu)^2 \quad (\text{Slide 38}).$$

$$\therefore \frac{\partial S(\underline{x}, \mu, \phi_1)}{\partial \phi_1} = -2 \sum_{t=2}^n (x_t - \mu)(x_{t-1} - \mu) + 2\phi_1 \sum_{t=2}^{n-1} (x_t - \mu)^2$$

$\therefore \ln (2)$

$$-\frac{\hat{\phi}_1}{1 - \hat{\phi}_1^2} - \frac{1}{2\hat{\sigma}_z^2} \left\{ 2\hat{\phi}_1 \sum_{t=2}^{n-1} (x_t - \hat{\mu})^2 - 2 \sum_{t=2}^n (x_t - \hat{\mu})(x_{t-1} - \hat{\mu}) \right\} = 0$$

$$\frac{\partial \ell}{\partial \sigma_z^2} \bigg|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0$$

$$\Rightarrow -\frac{n}{2\hat{\sigma}_z^2} + \frac{1}{2(\hat{\sigma}_z^2)^2} S(\underline{x}; \hat{\mu}, \hat{\phi}_1) = 0$$

$$\Rightarrow \hat{\sigma}_z^2 = \frac{S(\underline{x}; \hat{\mu}, \hat{\phi}_1)}{n}.$$